

Internship Report

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Introduction

Althion is a med-tech company developing IoT enabled innovative medical equipment and devices that solve pressing healthcare problems. The company works closely with healthcare practitioners in ideating, prototyping, testing and manufacturing affordable health care products. The product lineup includes cutting-edge RO and ultra-pure water systems, engineered to meet stringent medical and laboratory standards, ensuring patient safety and operational efficiency.

Following are few latest products used by various customers:

BW Series – 500–2000 LPH Double Pass RO Systems: IoT-Enabled Ultra-Pure Water Solutions for Hemodialysis Centers & Hospitals

AT250 – 100/250 LPH Portable RO Systems: Ultra-Pure Water for Home & ICU Dialysis

UPW-1 & UPW-2 Ultra-Pure Water Systems: ASTM Type-1 & Type-2 Water for Labs, Biotech & Pharma



Complete details of the company and products are available at <https://althion.in/>

Internship

The internship at Althion is focused on designing new components for 1500 LPH double pass RO system used in the dialysis. In addition, few existing components are redesigned and improved for better operational efficiency. **Autodesk Fusion** software is used extensively for designing the components, and the exported files are shared with the manufacturers for the production.

Components

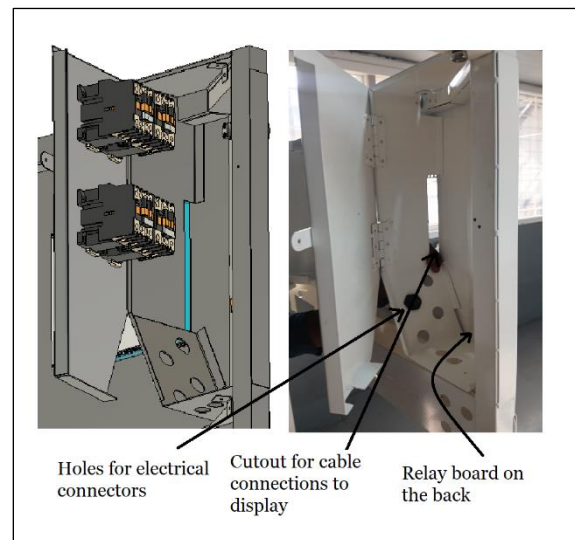
Following are various components that are designed or improved during the internship.

Contactor Box

The Contactor box is designed to hold four electrical power contactors for maintaining power supply to the motors for the full RO system (two pumps on the frame and another two elsewhere). The box also holds relay board that is designed to give electrical signals to the contactors.

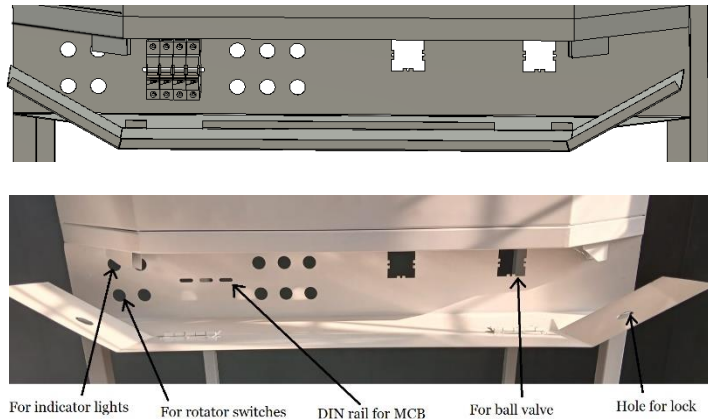
The contactor box is made from 1.2mm thick SS304 sheets and is power coated for long durability. Its dimensions are

11.5 cm x 17.6 cm x 42.5 cm with a door and simple locking system. The box is laser cut and machine modeled with 10mm bends for better strength. The design and finished product are shown in the above figure.



Control Panel

The control panel is the user interactive board holding various indicator lights and rotator switches that control different parts of the system. The board is made from 1.2mm thick SS304 sheet and laser cut

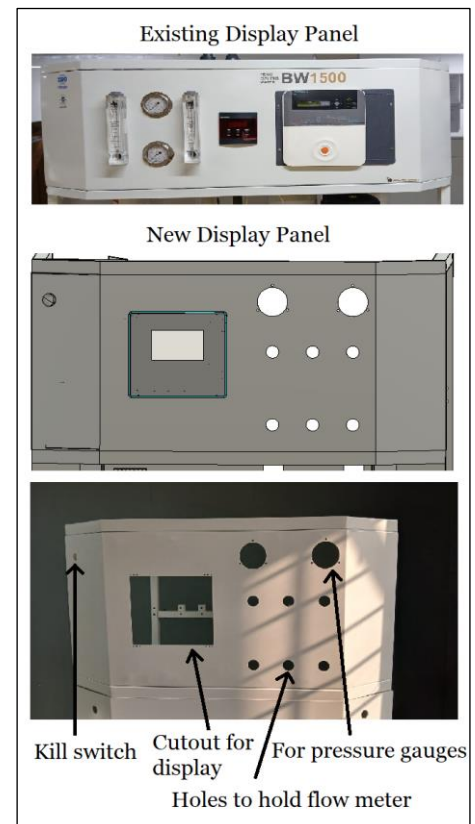


for precise and accurate shapes and has dimensions of 90 cm x 12.2 cm x 8.5 cm. It is designed for five pairs of indicator lights and rotator switches, two ball valves to control water flow and an MCB for control of overall power supply.

Display Panel

The display panel is designed to display various meter readings of the system to the user. The existing panel is redesigned to incorporate new features and for better user interface. The new design accommodates three flow meters, two pressure gauges and a digital display that is aesthetically designed for symmetry. A kill switch is added and can be used in case of emergencies to bypass the automated system.

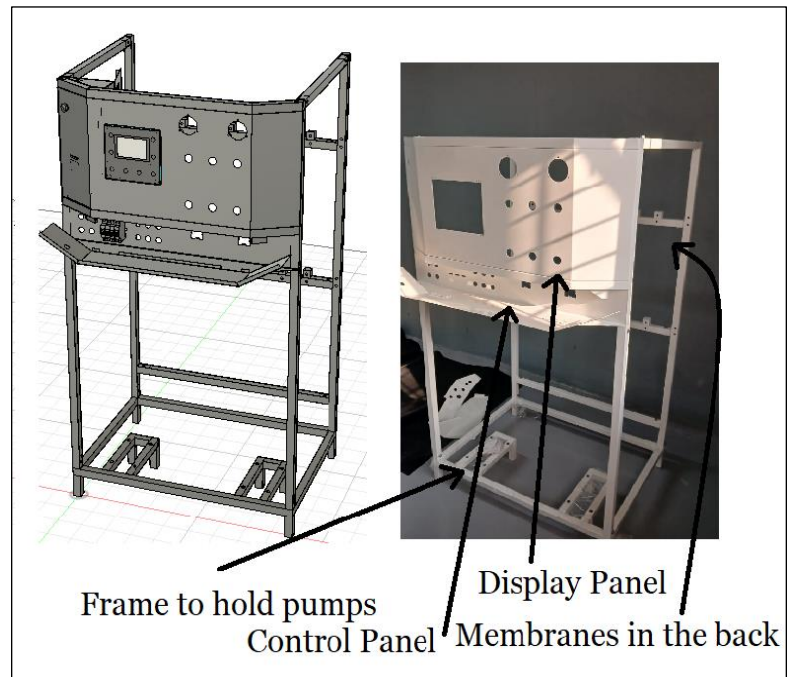
The panel is made from 1.2mm thick SS304 sheet with dimensions of 90 cm x 46 cm x 11.3 cm.



Structural Frame

The metal frame is the key structure that holds all components of the unit together. It is made from 1.2mm thick, 30mm x 30mm SS304 steel rods.

The frame is designed to hold a 2HP pump, a 4HP pump, an 80-40 1000 LPH membrane and two 40-40 250 LPH membranes. In addition, the contactor box, control panel and display panel are all fitted



together to the frame. The dimensions of the frame are 164 cm x 73.7 cm x 90.2 cm.

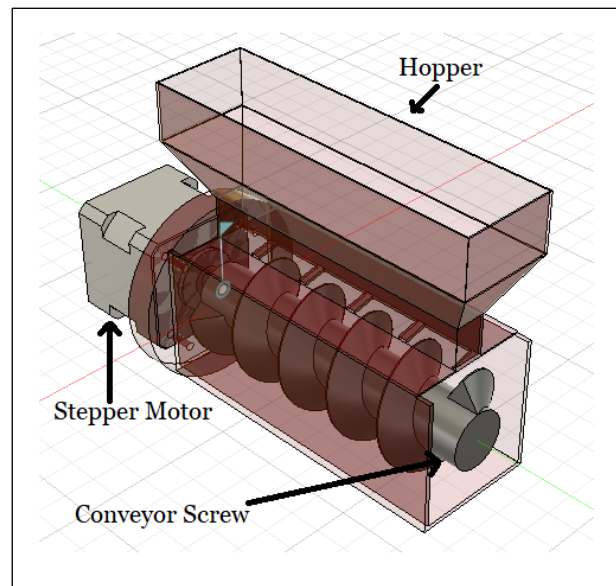
Prototype of Tablet Dispenser for Urea Clearance Monitor

A urea clearance monitor is a device used to assess the urea concentration in spent dialysate following a dialysis session. The system includes both inlet and outlet fluid lines, each equipped with a flow meter and a total dissolved solids (TDS) or conductivity sensor for monitoring flow conditions and solute levels. Each line is driven by a bidirectional peristaltic pump that routes fluid to a thermos-style mixing vessel containing a magnetic stirring element to ensure uniform mixing.

When a difference is detected between the TDS readings, indicating the onset of dialysis, the inlet line is used to flush and clean the thermos. Subsequently,

the outlet peristaltic pump delivers spent dialysate into the thermos at a controlled, low flow rate. At the end of the dialysis session, a reagent tablet is released into the thermos, where it dissolves in the collected mixture. The resulting solution is then analyzed using a conductivity sensor. After measurement, the peristaltic pump transfers the mixture back to the outlet line for disposal.

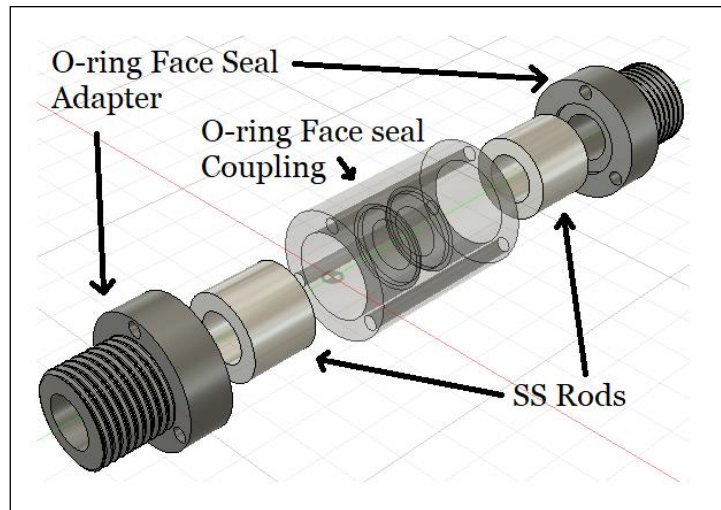
A prototype tablet dispensing mechanism is also incorporated into the system. The dispenser is designed to release a single tablet per operation. Tablets are stored in a hopper and metered by a helical conveyor screw, with each tablet occupying a single gap between adjacent threads. The screw is driven by a stepper motor, and a predefined angular rotation causes one tablet to be released into the thermos. The overall dimensions of the dispenser, including the stepper motor, are approximately 16 cm × 6.4 cm × 10 cm.



Prototype for Conductivity Sensor Housing

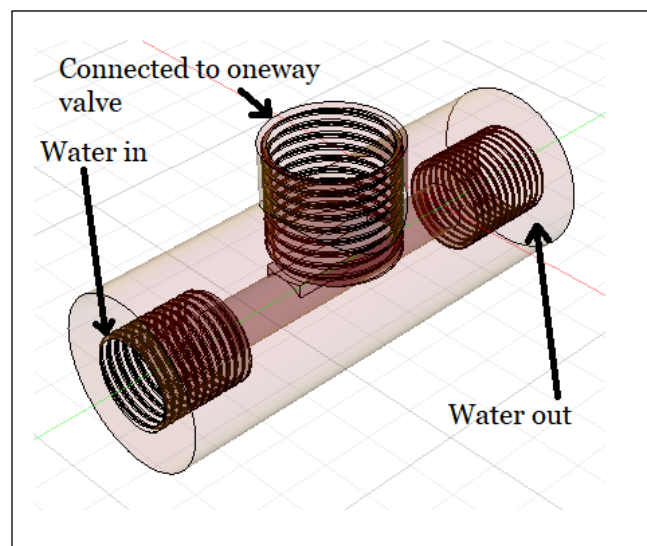
A conductivity sensor housing is the physical body that fits into a pipeline and allows liquid to flow through, while providing a defined space for conductivity measurement. It is designed to hold and accurately position sensing electrodes and seals so that they are consistently exposed to the flowing fluid.

The housing includes threaded connections for easy inline installation, along with O-rings to seal and prevent leakage while maintaining a stable internal flow path suitable for reliable measurements. The pipes are made with SS and the adapters along with the couple are 3D printed from resin. It has dimensions of 95mm length and 15mm diameter.



Venturi for RO water systems

A Venturi device is designed to generate suction within a flowing water stream, enabling the controlled entrainment of air to produce bubbles in the flow. The component incorporates two 1/8" threaded ports for inlet and outlet connections, allowing inline installation. A 5 mm internal channel is located at the throat of the Venturi and is connected to a threaded side port that interfaces



with an external one-way valve, permitting air to be drawn into the water stream while preventing reverse water flow. The overall dimensions of the device are approximately 50 mm in length, 23.5 mm in height, with a nominal diameter of 9 mm.